
46211 : Offshore Wind Energy Fall 2025
**Assignment 5: Dynamic model for a
simplified spar floater supporting the IEA
22MW wind turbine**

Prepared by

Sithik Aliyar and Henrik Bredmose
FLOWS section
Department of Wind and Energy Systems

Friday 14th November, 2025

Introduction

The fifth assignment concerns a dynamic model of a simplified spar floater supporting the IEA 22 MW reference wind turbine [2]. The assignment consists of three parts: model formulation, dynamic analysis and modification of the blade pitch controller.

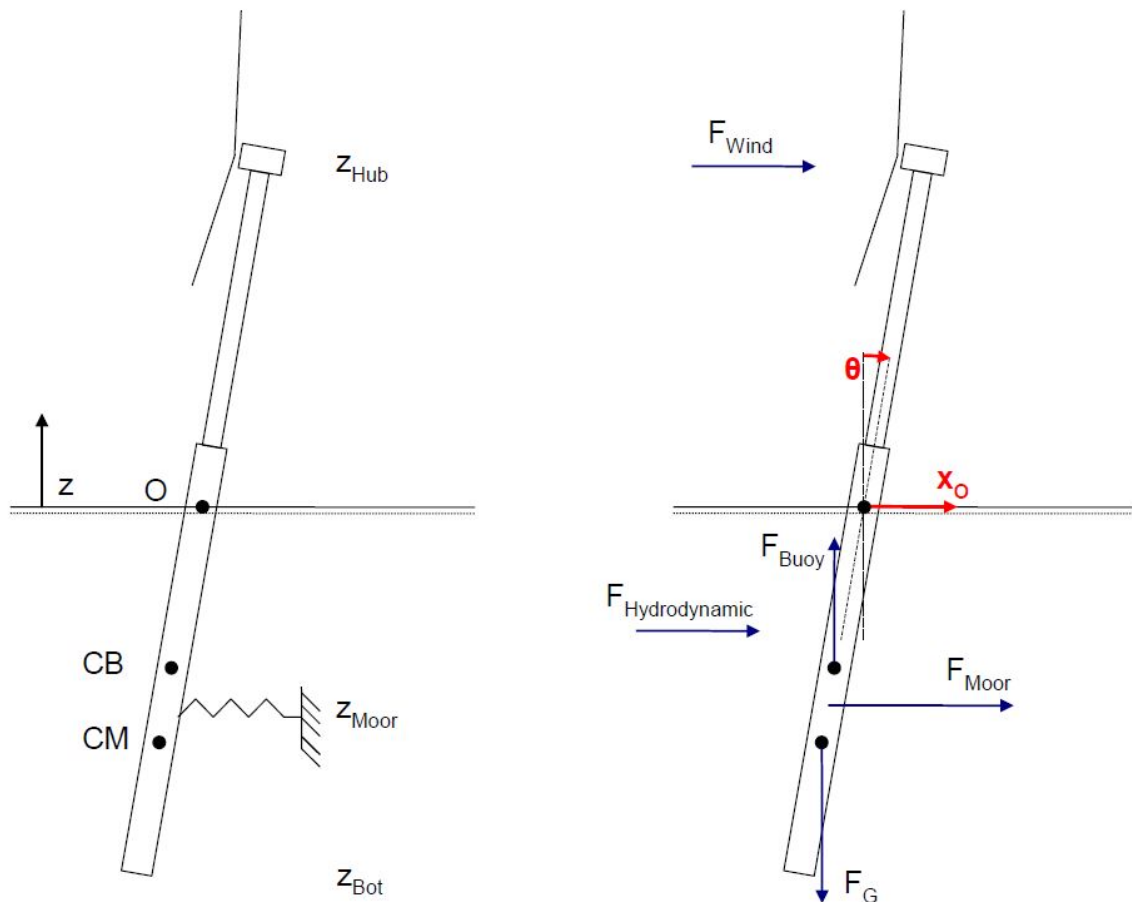


Figure 1: Definition sketch

Recommended time plan and report length

To ensure a proper progression and validation of the results, a recommended time plan is given below. We expect a maximum of 20 pages plus the code appendix.

- Part 1: To be done 14 November and the following days
- Part 2: To be done 18-21 November and the following days
- Part 3: To be done 25-28 November and the following days

Submission deadline: 1 December 2025 23:59

Part 1: Model formulation

The first part deals with the analytical formulation of the model. Consider the floating wind turbine in Figure 1. It consists of a spar buoy, a tower, a rotor and a nacelle (considered as turbine together), which are all rigidly connected. In this simplified model, 2D motion is assumed with 2 degrees of freedom: surge and pitch. The wind turbine is moored by a simple linear spring, which supplies a horizontal mooring force at the attachment point whenever its horizontal position differs from the resting position. The wind turbine is subject to an external wind load acting horizontally at the hub height, and the spar is subject to hydrodynamic forces from linear waves, added mass and damping. The parameters defining the model are given below.

$$\begin{aligned}
 M_{\text{Floater}} &= 5.6832e + 06 \text{ kg} \\
 z_{\text{CM,Floater}} &= -42.9013 \text{ m} \\
 M_{\text{Ballast}} &= 2.4000e + 07 \text{ kg} \\
 \text{Height of Ballast} &= 5 (\% \text{ of draft}) \\
 I_{\text{CM,Floater}} &= 6.3539e + 09 \text{ kgm}^2 \\
 M_{\text{Tower}} &= 1.5616e + 06 \text{ kg} \\
 z_{\text{CM,Tower}} &= 76.17 \text{ m} \\
 I_{\text{CM,Tower}} &= 2.6226e + 09 \text{ kgm}^2 \\
 M_{\text{Turbine}} &= 1215000 \text{ kg} \\
 z_{\text{Hub}} &= 170.00 \text{ m} \\
 D_{\text{Spar}} &= 20.00 \text{ m} \\
 \text{draft or } z_{\text{Bot}} &= -100.80 \text{ m} \\
 C_m &= 1.0 \\
 C_D &= 0.6 \\
 \rho_{\text{Water}} &= 1025 \text{ kg/m}^3 \\
 g &= 9.81 \text{ m/s}^2 \\
 h &= 320.00 \text{ m} \\
 K_{\text{Moor}} &= 150075 \text{ N/m} \\
 z_{\text{Moor}} &= -50.40 \text{ m}
 \end{aligned}$$

A dynamic model for the surge and pitch motion at the point of flotation (O) is given by:

$$M \begin{bmatrix} \ddot{x}_0 \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F_{\text{Moor}} + \tilde{F}_{\text{Hydrodynamic}} + F_{\text{Wind}} \\ \tau_{\text{Buoy}} + \tau_{\text{Moor}} + \tilde{\tau}_{\text{Hydrodynamic}} + \tau_{\text{Wind}} \end{bmatrix}. \quad (1)$$

In this model, the surge and pitch deflections are assumed small, allowing for small-angle approximations for θ .

1. Calculate the total mass M_{Tot} , the position of the center of mass $z_{\text{CM,Tot}}$, and the moment of inertia around point O , $I_{O,\text{Tot}}$, for the entire structure.
2. Using a sketch, express the local surge displacement $x(z)$ as a function of (x_0, θ) .

Table 1: IEA 22 MW reference wind turbine properties

Property	Value
Power Rating	22 MW
Turbine Class	1B
Cut-in Wind Speed (m/s)	3
Rated Wind Speed (m/s)	11
Cut-out Wind Speed (m/s)	25
Minimum Rotor Speed (rpm)	1.8
Maximum Rotor Speed (rpm)	7.06
Maximum Tip Speed (m/s)	105
Rotor Diameter (m)	284
Hub Height (m)	170

- Express F_{Moor} and τ_{Moor} as functions of (x_0, θ) .
- Express the restoring moment from gravity and hydrostatic forces, τ_{Buoy} , as a function of (x_0, θ) .
- Express the integrated hydrodynamic force and moment $\tilde{F}_{\text{Hydrodynamic}}$ and $\tilde{\tau}_{\text{Hydrodynamic}}$ in terms of (x_0, θ) and their time derivatives, given that the horizontal hydrodynamic force for a slice of the spar buoy of height $\tilde{z}$ is

$$d\tilde{F}_{\text{Hydrodynamic}} = \rho_{\text{water}} C_m A (\dot{u}(z) - \ddot{x}(z)) dz + \rho_{\text{water}} A \dot{u}(z) dz + \frac{1}{2} \rho_{\text{water}} C_D D (u(z) - \dot{x}(z)) |u(z) - \dot{x}(z)| dz. \quad (2)$$

Here, A and D are the cross-sectional area and diameter of the spar buoy at the upright position.

- Reformulate the model in the following form:

$$\left(\begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} + \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \right) \begin{bmatrix} \ddot{x}_0 \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{bmatrix} x_0 \\ \theta \end{bmatrix} = \begin{bmatrix} F_{\text{Hydrodynamic}} + F_{\text{Wind}} \\ \tau_{\text{Hydrodynamic}} + \tau_{\text{Wind}} \end{bmatrix}. \quad (3)$$

Give analytical expressions and numerical values for all elements in M , A , and C . Further, give analytical expressions for all hydrodynamic forcing terms on the right-hand side.

Hint: The elements of M can be found from lecture slides and the answer to question 1. The elements of A and C can be derived from questions 3, 4, and 5.

- Using the homogeneous version of (3), estimate the natural frequencies for surge and pitch.

Part 2: Dynamic analysis

The second part focuses on the numerical implementation of the model and the analysis of its response to wind and wave conditions. The test runs will follow selected load cases as specified in [1].

General Hints: When calculating response time series to external loads, consider a total simulation length of 600 s plus an initial transient period of at least 600 s. Analysis (e.g., PSD, standard deviation) should exclude the transient period. Results should be presented with subplots where time series (left) and PSD (right) are shown for wind speed at the hub, free-surface elevation, floater surge, and floater pitch. For the wind and wave spectra, use a high cut-off frequency of $f_{\text{high cut}} = 0.5 \text{ Hz}$ and set the frequency resolution as $\Delta f = 1/T_{\text{Dur}}$, where T_{Dur} is the duration of the signal. For the ODE solver, specify $\text{tspan} = [0 : \Delta t : T_{\text{Dur}} - \Delta t]$ with $\Delta t = 0.1 \text{ s}$, and pre-compute the time series of wave kinematics and wind speed using a finer time step of $\Delta t = 0.05 \text{ s}$.

8. Re-write (Eq.(3)) to a first-order system by introducing the state vector

Define the vector $\mathbf{q}$ as:

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \begin{bmatrix} x_0 \\ \theta \\ \dot{x}_0 \\ \dot{\theta} \end{bmatrix}. \quad (4)$$

The first-order system must have the form

$$\frac{d\mathbf{q}}{dt} = \mathbf{f}(t, \mathbf{q}). \quad (5)$$

Express $\mathbf{f}(t, \mathbf{q})$ in terms of the quantities in Eq.(3).

9. Write a MATLAB program to solve Eq. (5) using the MATLAB ‘ode4’ solver. Initially, exclude the hydrodynamic and wind forcing terms. Introduce a linear damping term $B_{11}\dot{x}_0$ with $B_{11} = 6 \times 10^5 \text{ N/(m/s)}$ on the left-hand side of the surge equation in Eq. (3), and include this in the MATLAB code.
10. Solve the equations of motion for two decay tests in surge and pitch:

(a) First, use the initial condition $\mathbf{q} = [1 \ 0 \ 0 \ 0]^T$ and solve for a duration of 600 s.

(b) Next, set the initial condition to $\mathbf{q} = [0 \ 0.1 \ 0 \ 0]^T$.

Present time series and power spectral density (PSD) plots for surge and pitch in both tests. Analyze the results to confirm that the model exhibits the same natural frequencies as found in question 7. Determine if the surge and pitch motions are coupled.

11. Add hydrodynamic forcing terms, computed only up to the still water level (no Wheeler stretching is needed for this assignment).

- (a) Repeat the decay tests from question 10 and compare the time series results. Discuss any observed differences.
- (b) Perform a pitch decay test with the initial condition $\mathbf{q} = [0 \ 1 \ 0 \ 0]^T$ and hydrodynamic forcing terms disabled. Compare these results to those obtained with the hydrodynamic forcing terms enabled. Discuss whether the model remains reliable for this decay case.

12. Compute the response for a regular wave with height $H = 6$ m and period $T = 10$ s. Wave kinematics for the forcing can be evaluated at the upright position of the spar buoy. Present time series and PSD for surge and pitch and analyze their behavior. Note that the initial condition should be changed from the one associated with the decay tests.
13. Include wind forcing in the model, where the wind force at the hub is determined from a thrust coefficient C_T in three steps as follows:

1. First, calculate the mean thrust force as:

$$\bar{F}_{\text{wind}} = \frac{1}{2} \rho_{\text{air}} A_{\text{Rotor}} C_T (V_{10 \text{ min}}) V_{10 \text{ min}}^2. \quad (6)$$

2. Next, compute the time-varying thrust time series as:

$$\tilde{F}_{\text{wind}}(t) = \frac{1}{2} \rho_{\text{air}} A_{\text{Rotor}} C_T (V_{\text{rel}}) V_{\text{rel}} |V_{\text{rel}}|. \quad (7)$$

3. Finally, apply an ad-hoc reduction to the dynamic part to compensate for the spatial variation of turbulence across the rotor:

$$F_{\text{wind}}(t) = \bar{F}_{\text{wind}} + f_{\text{red}}(V_{10 \text{ min}}) (\tilde{F}_{\text{wind}} - \bar{F}_{\text{wind}}). \quad (8)$$

where:

$$V_{\text{rel}} = V_{\text{hub}} - \dot{x}_{\text{hub}}$$

and

$$C_T = \begin{cases} C_{T0}, & V < V_1, \\ C_{T0} - c(V - V_1), & V_1 \leq V < V_{\text{rated}}, \\ C_{T1} \exp \left[-a \frac{(V - V_{\text{rated}})^b}{(10 + V - V_{\text{rated}})^b} \right], & V \geq V_{\text{rated}}. \end{cases}$$

$$C_{T0} = 0.8,$$

$$C_{T1} = 0.582,$$

$$a = 4.644,$$

$$b = 1.199,$$

$$c = 0.109,$$

$$V_{\text{rated}} = 11.0 \text{ m/s},$$

$$V_1 = 9.0 \text{ m/s},$$

$$\rho_{\text{air}} = 1.22 \text{ kg/m}^3,$$

$$D_{\text{rotor}} = 284 \text{ m}.$$

The function $f_{\text{red}}(V_{10 \text{ min}})$ is defined as:

$$f_{\text{red}}(V_{10 \text{ min}}) = \begin{cases} 0.54 & V_{10 \text{ min}} < V_{\text{Rated}} \\ 0.54 + 0.027(V_{10 \text{ min}} - V_{\text{Rated}}) & V_{10 \text{ min}} \geq V_{\text{Rated}} \end{cases}$$

The numerical constants are derived from the IEA 22MW reference wind turbine specifications [2].

Repeat the results from question 12, adding a steady wind of 8 m/s at hub height. Compare these results to those from question 12, analyzing the effect of steady wind on the standard deviation (std) of wave-induced motions and the transient response.

14. Repeat the calculations from question 13, but this time with irregular waves (JONSWAP spectrum with $H_s = 6$ m, $T_p = 10$ s, $\gamma = 3.3$). Present the results as time series and PSD for surge and pitch. Analyse the power spectra: which motions are induced by the wind and which are induced by the waves? What causes the spectral peaks observed?
15. Extend the wind forcing to include a Kaimal wind spectrum. The following formulation of the spectral density function for the fluctuating part of the wind speed can be used:

$$S_{\text{Wind}} = \frac{4I^2 V_{10\min} l}{\left(1 + 6 \frac{fl}{V_{10\min}}\right)^{5/3}}. \quad (10)$$

Compute the motion of the wind turbine subject to irregular waves (JONSWAP spectrum with $H_s = 6$ m, $T_p = 10$ s, $\gamma = 3.3$) and turbulent wind (Kaimal wind climate of $V_{10\min} = 8$ m/s, $I = 0.14$, $l = 340.2$ m). Present the results as time series and PSD for surge and pitch. Analyse the power spectra: which motions are induced by the wind and which are induced by the waves? What causes the spectral peaks observed?

Part 3: Adaption of pitch control for dynamic stability

The third part consist of adapting the controller to the floating configuration and investigating the dynamic stability.

16. Consider a load case of steady wind with no turbulence and no waves. Compute the response for (a) $V_{10\min} = 10$ m/s and (b) $V_{10\min} = 16$ m/s. Explain why the floater pitch motion is different for the two cases.
17. For a real wind turbine, the adjustment of C_T to V_{rel} happens through dynamic control of the blade pitch. As this involves mechanical motion, the adjustment is associated with a time lag. In the present simplified model, the process can be represented by the following differential equation for C_T , where γ is a constant:

$$\frac{dC_T}{dt} = -\gamma(C_T - C_T(V_{\text{rel}})). \quad (11)$$

Include C_T as a fifth element of the state vector q and include (11) in the dynamic model. Demonstrate that for $\gamma = 2$, the new model gives similar results as in question 16(b).

18. Experiment with γ and suggest a value that makes the floater pitch response stable. What would this γ parameter represent in a real wind turbine? What is the consequence of choosing γ too small? Hint: think not only in terms of floater pitch stability.

References

- [1] Jason Jonkman and Walter Musial. *Offshore code comparison collaboration (OC3) for IEA Wind Task 23 offshore wind technology and deployment*. Tech. rep. National Renewable Energy Lab.(NREL), Golden, CO (United States), 2010.
- [2] Frederik Zahle et al. *Definition of the IEA Wind 22-Megawatt Offshore Reference Wind Turbine*. English. DTU Wind Energy Report E-0243 IEA Wind TCP Task 55. Technical University of Denmark, 2024. DOI: 10.11581/DTU.00000317.